The seven shared group structures in the hexagonal tables

A structural supplement to the accepted marked affine models

We recover the group-column assertions in Table 6, rows 2–5, and Table 8, rows 1–3, of Fernandes–Leemans–Ivić Weiss [1]. The original constructions and table assertions remain theirs. The complete earlier marked-kernel common theory and the paid *Rotation certificates and affine period families* supplement remain premises. In particular, the universal presentations, marked residues, intersection property and flag geometry are already proved there. Here we use those exact models to supply previously missing normal factors and splittings. We do not rerun their presentation or geometry computations.

The source adopts Atlas notation in its introduction. A colon records a split extension and a cross a direct product. Iterated extensions must be interpreted at the indicated stages: a complement at each stage does not imply a complement to the innermost kernel in the entire group. The actions in the source abbreviations are unnamed. Our realizations specify them by the accepted faithful affine actions and, for the fixed finite cases, explicit permutation generators and bijective maps. An unspecified action never replaces a proof of a kernel, quotient or splitting assertion.

1 One splitting for every cubic parameter

Let Q be the signed permutation group on three coordinates and let $\epsilon(q) \in \{0,1\}$ be the parity of its number of sign changes. The accepted cubic model is

$$D = \{v \in \mathbb{Z}^3 : \textstyle\sum v_i \equiv 0 \pmod 2\}, \qquad E = \{(q,v) \in Q \ltimes \mathbb{Z}^3 : \textstyle\sum v_i \equiv \epsilon(q) \pmod 2\}.$$

Products act on column vectors, with $(q,v)(r,w) = (qr, v+qw)$. The literal full group at parameter $s \geq 3$ is $G_s = E/sD$, of order $48s^3$. Its marking is the exact marking in the shared proof; no new isomorphism based merely on this order is used.

Lemma 1. *For every positive integer s, the abstract affine quotient has the split form*

$$E/sD \cong (D/sD) \rtimes Q.$$

For odd s this is $\mathrm{Dih}(2s) \wr S_3$*, in the imprimitive wreath action with three factors, where* $\mathrm{Dih}(2s)$ *has order* $2s$*.*

Proof. Put $p = (1/2, 1/2, 1/2)$. For every signed permutation q, the vector $p - qp$ is integral: its entries are zero or one, with a one exactly where the corresponding sign is negative. Thus $\sum_i (p-qp)_i \equiv \epsilon(q) \pmod 2$, and

$$\sigma(q) = (q, p - qp) \in E, \qquad \sigma(q)\sigma(r) = \sigma(qr).$$

The group $\sigma(Q)$ fixes p and meets the translation subgroup D trivially. An arbitrary $(q,v) \in E$ differs from $\sigma(q)$ by a translation whose coordinate sum is even. Hence $E = D \rtimes \sigma(Q)$. Quotienting by the invariant subgroup sD preserves this complement, proving the first assertion for every s.

If s is odd, reduction of coordinates gives a Q-equivariant isomorphism $D/sD \longrightarrow (\mathbb{Z}/s\mathbb{Z})^3$. Indeed, any coordinate residue has an even-sum lift after adding s to one coordinate if necessary. If $sw \in D$, oddness of s implies $w \in D$, proving injectivity. The three independent sign changes invert the three cyclic factors, and S_3 permutes them. This is precisely the displayed wreath product, including its action. □

It follows at once that Table 6 row 2 ($s = 3$) has

$$G_3 \cong (S_3 \times S_3 \times S_3) \rtimes S_3,$$

and row 4 ($s = 5$) has

$$G_5 \cong C_5^3 \rtimes (C_2^3 \rtimes S_3).$$

The three commuting sign factors refine the latter to the source's $5^3 : 2 : 2 : 2 : S_3$. For $s = 6$, the Chinese remainder isomorphism of the free lattice is Q-equivariant:

$$D/6D \cong D/2D \oplus D/3D.$$

The subgroup corresponding to the second summand together with the same $\sigma(Q)$ is a complement to $D/2D$. Consequently row 5 has

$$G_6 \cong C_2^3 \rtimes \big((S_3 \times S_3 \times S_3) \rtimes S_3\big).$$

This gives the successive split factors in the source's $2^3 : S_3 : S_3 : S_3 : S_3$. It makes no assertion that the action on the first C_2^3 is trivial.

2 The even cubic row and the fixed square group

For the remaining cubic parameter $s = 4$, write $T = D/4D$ and let $X \cong C_2^3$ be the three sign changes inside $\sigma(Q)$. Use the accepted faithful action on the $4s = 16$ plane labels. In that action set

$$N = \left\langle T^2,\ t_{(2,0,0)},\ X \right\rangle, \qquad C = \langle a, b, c \rangle,$$

where the accepted ordered marking is (a, b, d, c). These are explicit subgroups of the already identified full group G_4. The complete small certificate gives

$$|N| = 128, \quad Z(N) = N' = \Phi(N) \cong C_2, \quad N \trianglelefteq G_4, \qquad C \cong S_4, \quad N \cap C = 1, \quad NC = G_4.$$

Thus N is extraspecial, by its defining centre, derived and Frattini conditions. Inside the actual complement C, the normal Klein four group and a point stabilizer give $C = C_2^2 \rtimes S_3$. Table 6 row 3 therefore has the asserted structure

$$G_4 \cong 2^{1+6} : (2^2 : S_3).$$

The normality and complement identities, and the bijections to S_4 and S_3, are checked on these actual subgroups. They are not inferred from the orders 128, 24 and 6.

For the square family the accepted lattice and generators are

$$\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}h, \qquad h = \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4),$$

$$R_0(x) = (1 - x_1, x_2, x_3, x_4), \qquad R_1(x) = x - \tfrac{1}{2}(x_1 - x_2 - x_3 - x_4)(1, -1, -1, -1),$$

with R_2 negating x_4, R_3 interchanging x_3, x_4, and R_4 interchanging x_2, x_3. The earlier proof identifies $E_4 = \langle R_0, \ldots, R_4 \rangle$, its entire translation kernel Λ, and the source's full marked group as $E_4/s\Lambda$.

Theorem 2. *For every $s \geq 1$, Table 8 row 1 has the split structure*

$$E_4/s\Lambda \cong C_s^4 \rtimes \big(2^{1+4} \rtimes (S_3 \times S_3)\big).$$

It therefore realizes the successive source factors $s^4 : 2^{1+4} : S_3 : S_3$ uniformly.

Proof. The subgroup $Q_0 = \langle R_1, R_2, R_3, R_4 \rangle$ fixes the origin. Its action on the following 48 roots is faithful, since they span the ambient vector space:

$$\{\pm e_i\} \ \cup \ \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\} \ \cup \ \{\pm e_i \pm e_j : i < j\}.$$

The fixed certificate constructs these four permutations from the literal displayed linear maps. It verifies $|Q_0| = 1152$, equal to the entire linear image already proved for E_4. Thus Q_0 maps onto that image and meets the translation kernel trivially; consequently $E_4 = \Lambda \rtimes Q_0$. Its complement still injects after quotienting by every $s\Lambda$, including $s = 1$.

In this one fixed faithful Q_0, the certificate finds its normal 2-core N_0 and an actual complement C_0, and verifies

$$|N_0| = 32, \quad Z(N_0) = N_0' = \Phi(N_0) \cong C_2, \quad Q_0 = N_0 C_0, \quad N_0 \cap C_0 = 1.$$

An explicit bijective homomorphism identifies C_0 with $S_3 \times S_3$; the inverse images of the two factors commute, intersect trivially and generate C_0. Therefore N_0 is extraspecial and the fixed linear structure is proved. Combining it with the uniform affine splitting proves the theorem. No list of sampled values of s is used. □

3 The two inherited finite presentations

Let H_1, H_2 be the exact full marked groups of orders 720 and 46080 in the accepted L121 finite proof. Its complete 13- and 30-point models are copied byte for byte as `inputs/model_data.json`. Their universal orders and geometry remain the accepted premises. The new program uses only these faithful models for structural witnesses.

For H_1 it constructs the normal perfect residual A and centre Z, identifies $A \cong A_5$ bijectively, and verifies

$$|Z| = 2, \quad A \cap Z = 1, \quad AZ = A \times Z \trianglelefteq H_1.$$

It then gives a subgroup $C \cong S_3$ with $C \cap AZ = 1$ and $(AZ)C = H_1$. This proves the printed Table 8 row 2 structure $(A_5 \times 2) : S_3$.

For H_2, let P be its normal perfect residual and Z its centre. The complete witnesses are

$$|P| = 960, \quad |Z| = 2, \quad P \cap Z = 1, \quad P = V \rtimes A, \quad V \cong C_2^4, \quad A \cong A_5,$$
$$PZ = P \times Z \trianglelefteq H_2, \qquad H_2 = (P \times Z) \rtimes C, \qquad C \cong S_4.$$

The proof checks elementary abelianness of the actual V, normality, all complement intersections and products, and both named bijections. To make the source's mixed string precise, put $C = W \rtimes S$ with $W \cong C_2^2$ and $S \cong S_3$, using the usual Klein four and point stabilizer through the verified S_4 map. Choose two independent involutions u, v in W. The program verifies the successive identities

$$K_1 = P \rtimes \langle u \rangle, \qquad K_2 = K_1 \times Z, \qquad K_3 = K_2 \rtimes \langle v \rangle, \qquad H_2 = K_3 \rtimes S.$$

Their orders are $1920, 3840, 7680, 46080$, respectively. Together with $P = 2^4 : A_5$, this realizes Table 8 row 3 as $2^4 : A_5 : 2 \times 2 : 2 : S_3$, with every operation given its indicated stage. We do not assert that the whole group splits over V; that stronger assertion is not needed for this iterated structure.

4 The exact finite premise and its limits

The complete new mathematical programs are `prepare_models.py` and `verify.g`; `run.py` provides their portable replay. The first constructs the faithful cubic plane actions and square root action directly from the accepted coordinates and copies the accepted finite models. The second records all subgroup generators, named isomorphism images, quotient images, and complement data. Each split extension checks normality, trivial intersection and generation, and verifies that the complement maps bijectively onto the actual quotient. Each direct product checks commutation as well. Extraspecial notation is justified by equality of the centre, derived and Frattini subgroups of order two. GAP's exact finite permutation, normal-subgroup, complement and isomorphism algorithms are retained computational trust. A structure-description string is never a proof premise.

The checks involve the four cubic images, one fixed linear group, and two inherited finite groups only. They do not repeat the 66-row presentation census, the coset-geometry checks or the old Lean work. The uniform cubic and square assertions have the written lattice and splitting proofs above; the finite certificates establish only their fixed structural components. All seven rows preserve their accepted markings and geometry. No other printed group column, source question, whole-original admission, numerical factor or new formalization follows from this supplement.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition, Tables 6 and 8, physical pp. 15 and 18.

[2] *Rotation certificates and affine period families*, complete accepted marked-model supplement, sections 4–15; the earlier *Supported kernels, translation lattices and finite marked quotients* remains fully acquired.

[3] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, retained twelve-page author edition; exact corrected proofs and finite models as separately accepted and migrated.